

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Who was the world's first programmer? You might be surprised to learn her name!



Learning Outcome

Describe social, urban and environmental change in written and spoken situations according to work and social contexts.

Ask your partner:

- What is the most challenging field that you can study at a university?
- In your opinion, in which field has enrollment increased the most over the past decade?

Can Do Statements

- I can talk about fame, work, and social or urban change.
- I can talk about what is being done or what has been done.
- I can speculate about the past.





Let's Explore!

Evidence suggests Ada Lovelace may have written the first **algorithm***.

* An "algorithm" is a set of rules which is usually carried out by a computer.



Instructions: Read the text aloud.

Although efforts are being made to change the perception, mathematics and computer programming have typically been seen as male disciplines. This impression might not have become so prevalent if more people knew that the first programmer may have been a woman.

Augusta Ada King, Countess of Lovelace, was born on December 10, 1815. From an early age, she was fascinated by mathematics and logic. Her talent in math led her to collaborate with Charles Babbage, who was a philosopher, inventor and mechanical engineer. He created plans for an "Analytical Engine" (a type of computer). Augusta had made notes on the proposal for the computer. These notes have been cited as the first algorithm intended to be used to make a machine function. Because of this, Augusta has been credited with writing the first computer program, and with increasing frequency she is acknowledged for being the first computer programmer. It must have been quite difficult for her, working in such a male-dominated society. Rumors of scandal, due to her informal relationships with men, persisted throughout her life.

Nowadays, more and more women are entering computer science and math, and groups such as "Women in Mathematics" and "Women in Computing" are being founded by student organizations. The advent of these groups, along with the trail that was blazed by women such as Ada Lovelace, could have been responsible for the increase in female enrollment that is currently being experienced by universities around the world.





Let's Practice!

A working vocabulary

Instructions: Underline the incorrect words in the sentence. Write the sentences correctly.

1. Mathematics and computer programming have typically been seen as male discipline.

2. Her mathematical talents led her to collaborates with Charles Babbage.

3. There were many rumor about her relationship.

4. These rumors caused a huge scandalous in London.

5. The famous inventer has an exciting new idea.



Let's Practice!

What steps are being taken to increase enrolment?

Instructions: Use the words from the box to correctly complete the sentences. Read the sentences aloud. Check your pronunciation with your instructor.

Word Bank

are being founded by / are being made / is currently being experienced by / has been credited with / have typically been seen as

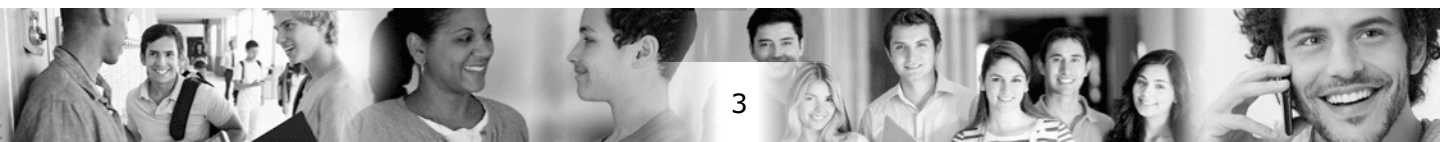
1. Efforts _____ to change the perception.

2. Mathematics and computer programming _____ male disciplines.

3. As a result, she _____ writing the first computer program.

4. Groups such as "Women in Mathematics" and "Women in Computing" _____ student organizations.

5. An increase in female enrolment _____ universities around the world.





Extension Activity

Talk about social and environmental changes that have been occurring in your country.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Create a chart with two columns.
- Add the headings "social change" and "environmental change."
- Write at least 5 examples in each column. (e.g. *Many employees are being replaced by robots.*)
- Compare your chart with a partner's.

Self-Assessment

Yes No

I create a chart comparing social and environmental change in my country.

☐ ☐

I use sentences in the passive voice.

☐ ☐

I share my thoughts with a partner.

☐ ☐

I provide and receive feedback on my ideas about social and environmental change.

☐ ☐


Target Words

- **discipline** (noun): a branch of knowledge studied at a college or university
e.g. Mathematics and computer programming have typically been seen as male disciplines.
- **programmer** (noun): a person who writes computer programs
e.g. The first programmer may have been a woman.
- **to collaborate** (verb): to work together on an activity
e.g. Her mathematical talents led her to collaborate with Charles Babbage.
- **philosopher** (noun): a person who studies existence, reality, and truth
e.g. Socrates was a Greek philosopher.
- **inventor** (noun): a person who creates new devices as an occupation
e.g. Thomas Edison was a famous American inventor.
- **mechanical engineer** (noun): an engineer who works with mechanical systems
e.g. They called a mechanical engineer to fix the engine.
- **rumor** (noun): a story which is uncertain or untrue
e.g. Many people spread rumors about her relationships.
- **scandal** (noun): public outrage caused by an illegal or immoral action
e.g. She always seemed to be at the center of some sort of scandal.





Learning Points

The Passive: The Present Continuous and Present Perfect Tenses

You use the Present Continuous to talk about an action that is happening at the time of speaking. The action or event is in progress.

*Student organizations **are founding** groups such as "Women in Mathematics" and "Women in Computing".*

You use the Present Perfect to talk about a finished action or event with present consequences. There is a connection between what happened in the past and the present.

*People **have** typically **seen** mathematics and computer programming as male disciplines.*

You often choose the passive form when it is not important to say who or what did the action. Passives without *agents* (the doers) are common in formal writing.

Look at the chart below for the structure of the passive voice.

Name	Structure	Example
Present Continuous	am/are/is + being + past participle	<i>Groups such as "Women in Mathematics" and "Women in Computing" are being founded by student organizations.</i>
Present Perfect	have/has + been + past participle	<i>Mathematics and computer programming have typically been seen as male disciplines.</i>

Examples:

- *Although efforts are being made to change the perception, mathematics and computer programming have typically been seen as male disciplines.*
- *These notes have been cited as the first algorithm intended to be used to make a machine function.*
- *Because of this, Augusta has been credited with writing the first computer program, and with increasing frequency she is acknowledged for being the first computer programmer.*
- *Nowadays, more and more women are entering computer science and math, and groups such as "Women in Mathematics" and "Women in Computing" are being founded by student organizations.*





Learning Points

Speculation About the Past

You can use modals to talk about degrees of certainty and speculate about things in the past.

You use them to describe what is definite, probable, possible, or impossible from the speaker's point of view.

You can use *must*, *may*, *might*, and *could* to describe degrees of certainty.

You can use modals with the Present Perfect to talk about the possibility of a past event that is not clear in the present context.

Example:

*This impression **might not have become** so prevalent if more people knew that the first programmer **may have been** a woman.*

Look at the chart for the meanings of modals when speculating about the past:

Modal Phrase	Meaning
Must have	It is very likely.
May have Might have Could have	It is possible.

- *May have* has a slightly stronger meaning than *might have* or *could have*.
- You can add **not** to make a negative speculation (It **may not have been** true).

Examples:

- *It must have been quite difficult for her, working in such a male-dominated society, and rumors of scandal due to her relaxed relationship with men persisted throughout her life.*
- *The advent of these groups, along with the trail that was blazed by women such as Ada Lovelace, could have been responsible for the increase in female enrollment that is currently being experienced by universities around the world.*